Exploring Multi-Timestep Multi-Stage Diffusion Features for Hyperspectral Image Classification

Jingyi Zhou, *Student Member, IEEE*, Jiamu Sheng, Peng Ye, Jiayuan Fan, *Member, IEEE*, Tong He, Bin Wang, *Senior Member, IEEE*, and Tao Chen, *Senior Member, IEEE*

***Abstract*— The effectiveness of spectral–spatial feature learning is crucial for the hyperspectral image (HSI) classification task. Diffusion models, as a new class of groundbreaking generative models, have the ability to learn both contextual semantics and textual details from the distinct timestep dimension, enabling the modeling of complex spectral–spatial relations in HSIs. However, existing diffusion-based HSI classification methods only utilize manually selected single-timestep single-stage features, limiting the full exploration and exploitation of rich contextual semantics and textual information hidden in the diffusion model. To address this issue, we propose a novel diffusion-based feature learning framework that explores multi-timestep multi-stage diffusion features for HSI classification for the first time, called MTMSD. Specifically, the diffusion model is first pretrained with unlabeled HSI patches to mine the connotation of unlabeled data, and then is used to extract the multi-timestep multi-stage diffusion features. To effectively and efficiently leverage multi-timestep multi-stage features, two strategies are further developed. One strategy is class and timestep-oriented multi-stage feature purification (CTMSFP) module with the inter-class and inter-timestep prior for reducing the redundancy of multi-stage features and alleviating memory constraints. The other one is selective timestep feature fusion module with the guidance of global features to adaptively select different timestep features for integrating texture and semantics. Both strategies facilitate the generality and adaptability of the MTMSD framework for diverse patterns of different HSI data. Extensive experiments are conducted on four public HSI datasets, and the results demonstrate that our method outperforms state-of-the-art methods for HSI classification, especially on the challenging Houston 2018 dataset. The codes are available at https://github.com/zjyaccount/MTMSD.**

***Index Terms*— Denoising diffusion probabilistic model (DDPM), feature purification, feature selection, hyperspectral image (HSI) classification, multi-timestep multi-stage features.**

Manuscript received 30 December 2023; revised 26 April 2024; accepted 21 May 2024. Date of publication 30 May 2024; date of current version 10 June 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022ZD0160300, in part by the National Natural Science Foundation of China under Grant 62101137 and Grant 62071127, and in part by Shanghai Natural Science Foundation under Grant 23ZR1402900. *(Jingyi Zhou and Jiamu Sheng contributed equally to this work.) (Corresponding author: Jiayuan Fan.)*

Jingyi Zhou, Peng Ye, Bin Wang, and Tao Chen are with the School of Information Science and Technology, Fudan University, Shanghai 200433, China (e-mail: zhoujingyi19@fudan.edu.cn; eetchen@fudan.edu.cn).

Jiamu Sheng and Jiayuan Fan are with the Academy for Engineering and Technology, Fudan University, Shanghai 200433, China (e-mail: jmsheng22@m.fudan.edu.cn; jyfan@fudan.edu.cn).

Tong He is with Shanghai Artificial Intelligence Laboratory, Shanghai 200232, China.

Digital Object Identifier 10.1109/TGRS.2024.3407206

I. Introduction

HYPERSPECTRAL image (HSI) classification plays a crucial role in remote sensing, as it aims to distinguish each pixel's category in hyperspectral data by using dense and detailed electromagnetic spectral information [1], [2]. HSI classification has a wide range of applications in environmental monitoring [3], resource management [4], agriculture disaster response [5], military defense [6], etc.

The extraction of spectral–spatial features from HSIs plays a significant role in HSI classification tasks. Initially, machine learning-based feature extraction methods have been developed [7], [8], [9], [10], [11]. However, these approaches rely on manual feature engineering, resulting in limited extraction of discriminative information from the highly variable spectral data [12]. In recent years, the rapid advancement of deep learning has paved the way for neural network-based approaches for feature extraction, specifically leveraging convolutional networks (CNNs) [13], [14], [15] and transformer models [16], [17], [18], [19]. These neural network-based methods excel at automatically learning valuable spectral–spatial features from labeled HSI data, thus achieving promising results in HSI classification. Further, unsupervised methods such as [20], [21], and [22] have been designed to excavate spectral–spatial features from unlabeled HSIs. Typically, these methods employ an encoder–decoder network trained in an unsupervised manner for HSI reconstruction, thus extracting spectral–spatial features from unlabeled HSI data. Leveraging unsupervised feature learning enables deep mining of large amounts of unlabeled regions in HSI, which contain a wealth of label-agnostic information, thereby promoting the HSI classification task.

Recently, diffusion models [23], [24], [25] have emerged as powerful models with superior performance in generation and reconstruction tasks. These models have also been explored in various computer vision tasks such as semantic segmentation [26], [27], [28]. Distinguished from traditional deep neural networks, diffusion models employ a stepwise reverse denoising process, formulated as an iterative optimization procedure optimized by Langevin dynamics [29]. This approach introduces a multitude of degrees of freedom for feature learning and predicts additional information conditioned on the given noise-corrupted data at each timestep. Consequently, diffusion models can implicitly capture both high-level and low-level visual concepts, facilitating better generalization and modeling

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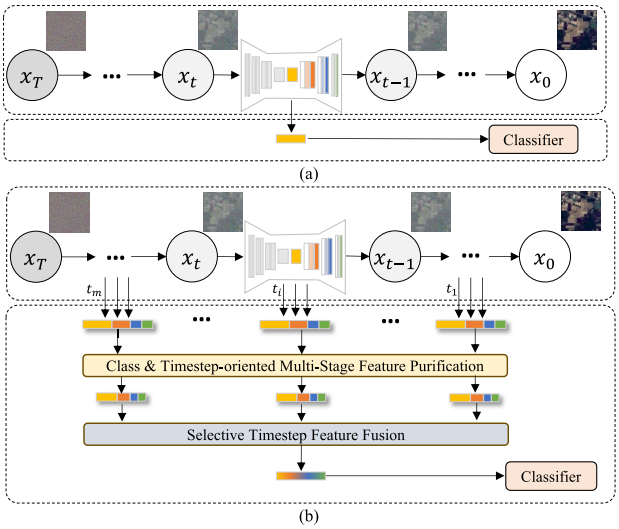


Fig. 1. Overview of existing diffusion-based feature learning frameworks for the HSI classification task. Our method can fully explore and exploit rich contextual semantics and textual features hidden in the diffusion model. (a) Existing method: manually selected single-timestep single-stage diffusion feature. (b) Our method: adaptively selected multi-timestep multi-stage diffusion feature.

of complex spectral–spatial relations [30], [31]. Therefore, one recent work has proposed using diffusion models for HSI classification [32], which employs unsupervised feature learning to extract diffusion features. However, the diffusion features used in [32] are extracted solely from a single timestep and a single stage of the denoising U-Net, and these selections about the timestep and stage are manually determined based on extensive experimentation with each dataset, as shown in Fig. 1. First, relying on single-timestep features from a single U-Net layer inevitably results in the loss of abundant spectral–spatial information and limits the effectiveness of modeling spectral–spatial relations. Second, the manual selection approach lacks generality and adaptability to diverse HSI datasets exhibiting specific spectral representation patterns.

Naturally, it is crucial to explore how to exploit abundant multi-timestep multi-stage features extracted from the whole stage of denoising U-Net more effectively and efficiently. First, multi-timestep features extracted from diffusion models are diverse and focus on different information. The shallow timestep-wise features are more informative for textual details, while deeper ones are more concerned about high-level semantics and global information [23], [33]. For modeling complex spectral–spatial relations, textual features depict the spatial distribution and changing patterns of ground objects, and contextual semantics represent the spectral attributes and the content of ground objects. Thus, leveraging multi-timestep features can integrate both contextual semantics and textual features to build comprehensive spectral–spatial representation for better performance. Second, multi-stage features, as compared to single-stage features, encapsulate a rich hierarchy of information that contains richer semantic and reconstruction characteristics, facilitating the modeling of spectral–spatial relations. Moreover, Chen et al. [32] also demonstrate that various stage features improve classification performance with different degrees across various datasets. Although multi-timestep multi-stage features are desired, HSIs from various datasets acquired by different sensors exhibit distinct spectral–spatial characteristics, leading to variations in the spectral–spatial representations. Moreover, for different regions in the same HSI, the emphasis on texture and semantics varies, leading to distinct preferences in the selection of timestep t. Meanwhile, multi-stage features are numerous and comprise both redundant semantics and reconstruction information along the channel dimension, posing a challenge in terms of memory constraints when training and inferring on large datasets.

In view of these, we propose a novel diffusion-based feature learning framework that explores multi-timestep multi-stage diffusion features for HSI classification for the first time, named MTMSD. More specifically, the proposed framework first pretrains a diffusion model with unlabeled HSI patches for diffusion feature learning, mining the connotation of unlabeled data that reveals the complex spectral–spatial dependencies. Then, we extract multi-timestep multi-stage diffusion features from the pretrained denoising U-Net decoder and construct the timestep-wise center and global feature bank by center extraction and average pooling. After that, two strategies are further developed in MTMSD to leverage multi-timestep multi-stage diffusion features effectively and efficiently. First, to reduce the redundancy of multi-stage features and maintain efficiency, we propose to perform class and timestep-oriented multi-stage feature purification (CTMSFP) on multi-stage features in the timestep-wise center feature bank with the inter-class and inter-timestep prior. Second, to effectively harness multi-timestep features and softly learn the proper timestep-wise feature combination for different datasets, we propose the selective timestep feature fusion module. This module is designed to adaptively select different timestep center features with the guidance of related global features, and fuse them for multi-timestep multi-stage selective representations that integrate contextual semantics and textual features to model comprehensive spectral–spatial relations. Ultimately, an ensemble of linear classifiers is employed for accurate HSI classification.

To summarize, our contributions are listed as follows.

1) For modeling complex spectral–spatial relations, we propose a novel diffusion-based framework that explores multi-timestep multi-stage diffusion features for HSI classification. To the best of our knowledge, this is the first work to learn and exploit multi-timestep multi-stage diffusion features for diffusion-based HSI classification.
2) We design the novel CTMSFP module. It adaptively selects significant channels of multi-stage diffusion features from both inter-class and inter-timestep aspects to reduce redundant information and maintain computational efficiency.
3) We propose to perform selective timestep feature fusion on multi-timestep diffusion features. This module allows each labeled patch of different datasets to adaptively select different timestep center features with the guidance of related global features, and fuse them for comprehensive multi-timestep multi-stage selective representations that integrate contextual semantics and textual information.

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4) Compared with several state-of-the-art HSI classification methods, experimental results demonstrate that our proposed method achieves significant classification accuracy on four public HSI datasets, especially on the challenging Houston 2018 dataset.

The remainder of this article is organized as follows. Section II describes related work. In Section III, our proposed MTMSD is introduced in detail. Section IV conducts extensive experiments on four HSI datasets to demonstrate the effectiveness of the proposed method. Finally, some conclusions are drawn in Section V.

II. Related Work

A. HSI Classification

HSI classification is an important research topic in the area of remote sensing. Since HSI classification aims to distinguish each pixel's category in hyperspectral data using dense electromagnetic spectral information [1], a large number of handcrafted feature-based methods have been designed for HSI classification [7], [8], [9], [10], [11]. Several works adopt morphological profiles (MPs) for manually extracting spectral–spatial features from HSIs. They achieve good classification results using MPs as input vectors with a support vector machine classifier. Subspace-based learning, such as sparse representation and manifold learning, is another common feature extraction strategy for HSI classification. These methods transform the high-dimensional original space using a low-dimensional subspace representation to learn spectral–spatial information.

Due to the remarkable breakthroughs achieved by deep learning in various computer vision tasks, many progressive deep learning-based networks have been widely utilized for HSI classification methods. Among these, CNNs draw significant attention with their feature extraction capability to extract spatially structural information and locally contextual information and become mainstream in HSI classification [13], [14], [15]. Based on the spectral and spatial attention modules, Zhong et al. [14] embed a residual block into a sequential spectral–spatial feature learning network. This architecture not only mitigates the risk of overfitting but also enhances classification performance. However, CNNs have the challenge of modeling long-term spectral information dependencies. To address this defect, researchers explore the value of the transformer and widely leverage it in HSI classification [16], [17], [18], [19]. Hong et al. [16] consider the spectral sequence of neighboring bands and design a pure transformer-based SpectralFormer (SF) backbone network, representing sequence attributes of spectral signatures. Sun et al. [17] extract high-level semantic features by introducing a Gaussian weighted token module into the transformer architecture, achieving promising performance in both classification accuracy and computational complexity.

In addition to supervised feature learning, unsupervised feature learning aims to learn feature representations from the input data without any annotated information, providing a solution to the limited labeled samples of HSI datasets. Typically, the commonly used unsupervised feature learning methods in HSI classification are based on the encoder–decoder paradigm, where an autoencoder-like network encodes the input HSI patches into a purified feature and then reconstructs the feature to initial HSI data by a decoder network. Mou et al. [34] first designed a fully 2-D Conv–Deconv network in an end-to-end manner for unsupervised feature learning of HSI classification. Similarly, Mei et al. [20] design a 3-D convolutional autoencoder (3D-CAE) for unsupervised feature learning of HSI classification. To alleviate the insufficiency of geometric representation and exploit the multi-scale features, Zhang et al. [21] design a multi-scale CNN-based unsupervised feature learning framework, with two branches of decoder and clustering optimized by the error feedback of image reconstruction and pseudo-label classification.

More recently, with the rise of diffusion models, one recent work proposes a diffusion-based HSI classification method. Chen et al. [32] propose SpectralDiff that extracts the spectral–spatial diffusion features from spectral–spatial denoising network and directly feeds them into the attention-based classification network for classification. However, they only use a single timestep feature from a single stage of the denoising U-Net, which is manually selected according to extensive experiments on each dataset. This results in a lack of information to model spectral–spatial relations and poor robustness to diverse HSI datasets with diverse spectral characteristics. In our work, we propose a novel diffusion-based HSI classification framework that explores multi-timestep multi-stage diffusion features. Through multi-stage feature purification and selective timestep fusion, our MTMSD enables adaptive integration of both contextual semantics and textual features to model complex spectral–spatial relations and generality for diverse patterns of different HSI data.

B. Diffusion Models

Diffusion models are a class of probabilistic generative models that progressively inject a standard Gaussian noise, then learn a model to reverse this process for sample generation [23], [24], [25]. Current research on diffusion models is mostly based on three formulations: denoising diffusion probabilistic models (DDPMs) [23], score-based generative models [31], and stochastic differential equations [35]. Among them, DDPMs are the mainstream diffusion models, and a large number of recent works based on DDPMs have made DDPMs increasingly powerful in terms of generative quality and diversity over other generative models [36]. Meanwhile, DDPMs have been widely used in several applications, including super-resolution [37], inpainting [38], and point cloud generation [39]. Recently, Baranchuk et al. [26] proposes a simple diffusion-based semantic segmentation approach that exploits three-timestep multi-stage features with manually selected timesteps and also proves that diffusion features capture high-level semantic information for semantic segmentation. However, a fixed timestep set results in suboptimal and non-generic features for different datasets. Additionally, such a simple utilization of diffusion features for segmentation may lead to inefficiencies due to redundant information and

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ineffectiveness in building semantic representations from diffusion features. Differently, we focus on the HSI classification task, and dynamically select the timesteps and stages of multi-timestep multi-stage features ranging from low to high, integrating textural features and semantics to build comprehensive spectral–spatial representations.

C. Deep Feature Selection

Feature selection [40], an essential process in deep learning-based computer vision, plays a pivotal role in improving model performance. The goal of feature selection is to retain informative and refined features from the original features, thereby reducing dimensionality and computational complexity. For different images, the important features vary, and feature selection assists models in adapting to data from diverse domains, extracting significant and refined features. Early feature selection methods select features with input-dependent soft attention. Hu et al. [41] introduce a channel attention module named SENet to adaptively recalibrate channel-wise features by exploiting the inter-channel relations. Similar to channel attention, spatial attention method GENet [42] is designed to enhance a network's capacity for context information modeling via spatial masks. Building on these, Woo et al. [43] combine both channel and spatial attention, introducing the CBAM method. Additionally, feature selection on different kernel features is also a self-adaptive and effective mechanism. Li et al. [44] propose SKNet to select the features extracted from different convolutional kernels using softmax attention along the channel dimension. Different from the above methods, our framework is the first to use feature selection on multi-timestep multi-stage features for HSI classification. In detail, MTMSD creatively employs inter-class and inter-timestep priors for multi-stage feature purification in the channel dimension, and concurrently conducts feature selection on multi-timestep features with the guidance of global information.

III. Method

Our proposed MTMSD is a novel diffusion-based feature learning framework and aims to explore multi-timestep multi-stage diffusion features effectively and efficiently for modeling spectral–spatial relations comprehensively. The framework is shown in Fig. 2. In Sections III-A–III-D, we introduce the proposed MTMSD in detail.

A. Diffusion Feature Learning on Unlabeled HSI

1) Brief Review of DDPM: DDPMs are a class of likelihood-based models that reconstruct the distribution of training data via an encoder–decoder denoising model. The denoising model is trained to remove noise from the training data destructed by Gaussian noises step by step. These models consist of a forward noising process and a reverse denoising process. In the forward process, Gaussian noise is added to the original training data $x_0 \sim q(x_0)$ step by step over T time steps, which follows the Markovian process:

$$q(x_t|x_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}x_{t-1}, \beta_t I\right) \tag{1}$$

where $\mathcal{N}(.)$ is a Gaussian distribution, and the Gaussian variances $\{\beta_t\}_{t=0}^{T}$ that determines the noise schedule are either be learned or scheduled. The above formulation leads that an arbitrary noisy sample x_t for each timestep t is obtained directly from x_0

$$x_t = \sqrt{\overline{\alpha}_t}x_0 + \sqrt{(1-\overline{\alpha}_t)}\epsilon, \epsilon \sim \mathcal{N}(0, I) \tag{2}$$

where $\alpha_t = 1 - \beta_t$, and $\overline{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. Then, in the reverse process, DDPM also follows a Markovian process to denoise the noisy sample x_T to x_0 step by step. Under large T and small β_t, the reverse transitions probability is approximated as a Gaussian distribution and is predicted by a learned neural network as follows:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_\theta(x_t, t)) \tag{3}$$

where the reverse process is re-parameterized by estimating $\mu_\theta(x_t, t)$ and $\sigma_\theta(x_t, t)$. $\sigma_\theta(x_t, t)$ is set to $\sigma_t^2\mathbf{I}$, where σ_t^2 is not learned. In practice, rather than predicting $\mu_\theta(x_t, t)$ directly, predicting the noise ϵ in (2) via a U-Net works best, and the parameterization of $\mu_\theta(x_t, t)$ is derived as follows:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\overline{\alpha}_t}}\epsilon_\theta(x_t, t)\right). \tag{4}$$

The U-Net denoising model $\epsilon_\theta(x_t, t)$ is optimized by minimizing the following loss function:

$$\mathcal{L}(\theta) = E_{t,x_0,\epsilon}\left[\left(\epsilon - \epsilon_\theta\left(\sqrt{\overline{\alpha}_t}x_0\right) + \sqrt{1-\overline{\alpha}_t}\epsilon, t\right)^2\right]. \tag{5}$$

In our work, improved DDPM [25] is adopted and has been proven to bring some improvements to the above DDPM. In detail, learned variances $\sigma_\theta(x_t, t)$ and an improved cosine noise schedule proposed in [25] lead to enhanced distribution learning ability.

2) Unsupervised Hyperspectral Diffusion Pretraining: Using the training skills and optimization objectives mentioned above, the DDPM for diffusion feature learning is trained with unlabeled hyperspectral data. Before training, the HSI data is pre-processed by principal components analysis (PCA) and random patch cropping operation. Then, given an unlabeled patch $x_0 \in \mathcal{R}^{H\times H\times D}$, where H is the patch size, and D is the number of PCA components, we gradually add Gaussian noise to the unlabeled HSI patch according to the cosine variance schedule $\{\beta_t\}_{t=0}^{T}$ in the diffusion process, where T is the total number of timesteps. Then, in the reverse process, a denoising U-Net is trained to predict the noise added on x_{t-1} taking noisy patch x_t and timestep t as inputs. And the sample x_0 can be obtained from the noise patch x_t by the iterative denoising steps according to (3). In each step of the training process, the timestep t is randomly sampled from 0 to T. The U-Net denoising model $\epsilon_\theta(x_t, t)$ is optimized by minimizing (5). The parameters in the pretraining process, such as patch size, number of PCA components, and total pretraining steps, will be discussed in Section IV-F.

B. Multi-Timestep Multi-Stage Diffusion Feature Extraction

After diffusion feature learning on unlabeled HSI, there exist abundant multi-timestep multi-stage diffusion features

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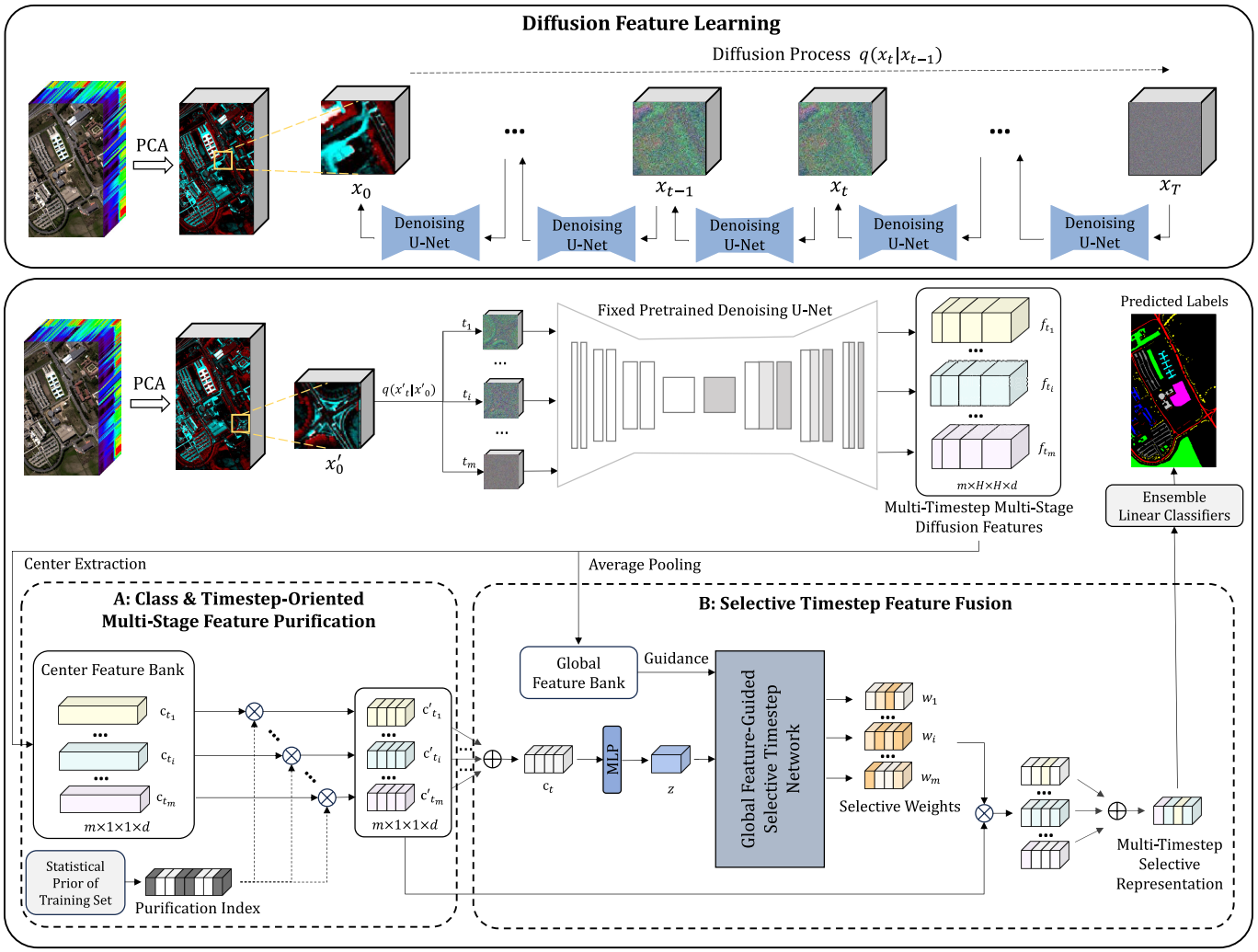


Fig. 2. Overview of our proposed MTMSD. The method consists of two steps. Step 1: We pretrain the DDPM with HSI patches in an unsupervised manner for diffusion feature learning. Step 2: We extract multi-timestep multi-stage diffusion features from the pretrained denoising U-Net decoder and construct the timestep-wise center and global feature bank by center extraction and average pooling. To effectively and efficiently leverage multi-timestep multi-stage features, we first perform class and timestep-oriented multi-stage purification on multi-stage features in the timestep-wise center feature bank, and then, we perform selective timestep feature fusion with global-feature guidance on the purified timestep-wise center feature bank. Classification is performed through an ensemble of lightweight classifiers.

that contain both contextual semantics and textual information hidden in the denoising U-Net with different timestep t. To model the complex spectral–spatial relations in HSIs, we extract multi-timestep multi-stage features from all stages of the fixed pretrained denoising U-Net decoder with all timesteps, and construct the timestep-wise center and global feature bank by center extraction and average pooling.

Specifically, given a labeled patch $x'_0 \in \mathcal{R}^{H\times H\times D}$ preprocessed by PCA to D channels, x'_0 is corrupted by adding Gaussian noise according to (2) and obtain $\{x'_{t_i}\}_{i=1}^m$ at a set of timesteps $\{t_i\}_{i=1}^m$ that are sampled from $[0, T]$ at equal intervals. Then, the noisy patches $\{x'_{t_i}\}_{i=1}^m$ are fed into the pretrained denoising U-Net to extract multi-timestep multi-stage diffusion features from all stages of the U-Net decoder. The different layer features are jointly upsampled to $H \times H$ and then concatenated to get the multi-stage feature $f_{t_i} \in \mathcal{R}^{H\times H\times d}$ at timestep t_i.

For each multi-stage feature f_{t_i}, we only reserve the center feature $c_{t_i} \in \mathcal{R}^{1\times 1\times d}$ located as $((H/2), (H/2))$ corresponding to the center pixel, and obtain the global feature $g_{t_i} \in \mathcal{R}^{1\times 1\times d}$ through global average pooling, which largely reduce computational cost with fewer memories. Following the above process, we construct the timestep-wise center feature bank $\mathcal{B}_c$ and the timestep-wise global feature bank $\mathcal{B}_g$:

$$\mathcal{B}_c = \left\{c_{t_i} \middle| i \in \{1, \ldots, m\}, c_{t_i} = \text{center}\left(f_{t_i}\right)\right\} \quad (6)$$

$$\mathcal{B}_g = \left\{g_{t_i} \middle| i \in \{1, \ldots, m\}, g_{t_i} = \text{avgpool}\left(f_{t_i}\right)\right\}. \quad (7)$$

C. Class and Timestep-Oriented Multi-Stage Feature Purification

Multi-stage features, extracted from the denoising U-Net, embody abundant reconstruction information from the pretraining process of the diffusion model. Despite this richness, the information is not entirely aligned with HSI classification requirements, containing redundant features irrelevant to the task. Furthermore, these features exhibit a degree of repetition among themselves. Therefore, the class and timestep-oriented multi-stage purification is proposed to explore multi-stage

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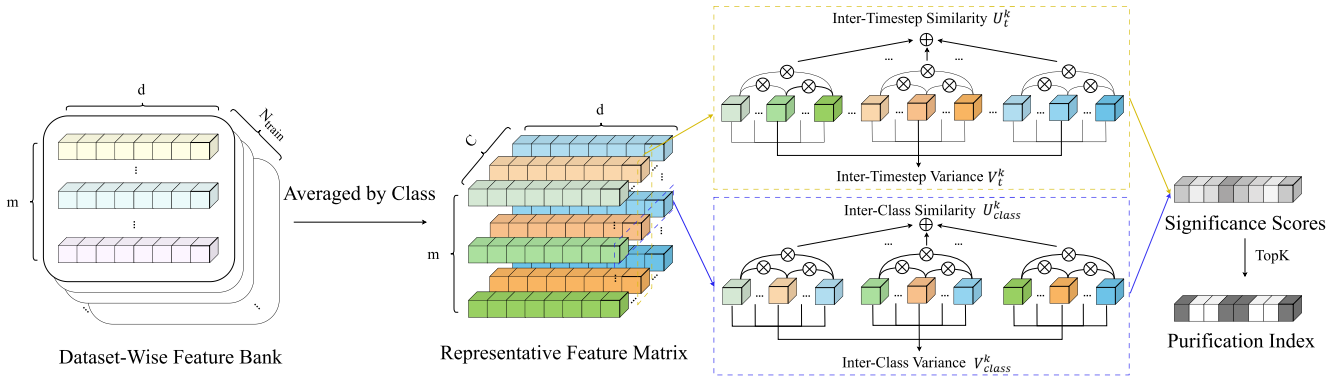


Fig. 3. Purification index generation of our proposed class and timestep-oriented feature purification module.

features by selecting significant channels, aiming to remove the redundant information and reduce the computational cost.

Before the multi-stage feature purification, we first generate the purification index using the prior of dataset-wise multi-timestep multi-stage feature bank, depicted in Fig. 3. Specifically, for a C-category HSI classification dataset, S^j is the training samples of category j, $j \in \{1, \ldots, C\}$. The feature banks of samples in S^j are averaged to get the category representative features of category j. Representing all training samples by the category representative features can significantly reduce computational overhead. Gathering all the representative features of each category, a representative feature matrix $M \in \mathcal{R}^{m \times C \times d}$ can be obtained

$$M_{i,j} = \frac{1}{|S^j|} \sum_{x \in S^j} c_{t_i}(x) \tag{8}$$

where $c_{t_i}(x)$ is the feature at t_i in the center feature bank of sample x, and $|S^j|$ is the size of S^j.

In order to purify the discriminative and effective channels for classification, a class-oriented significance score is proposed to evaluate the significance of each channel from inter-class relations. It aims to filter out channels that are highly homogenized and thus have little impact on classification by minimizing the inter-class similarity and maximizing the inter-class variance. For kth index, the class-oriented significance score τ_{class}^k is formed as

$$\tau_{\text{class}}^k = -\alpha U_{\text{class}}^k + (1-\alpha) V_{\text{class}}^k \tag{9}$$

where $\alpha \in (0, 1)$, U_{class}^k denotes the inter-class similarity at index k obtained by summing the average cosine similarities across classes at all the timesteps, and V_{class}^k denotes the inter-class variance at index k which is the sum of the variances across classes at all the timesteps, formed as follows:

$$U_{\text{class}}^k = \frac{1}{m}\frac{1}{c^2} \sum_{i=1}^{m} \sum_{\substack{p=1 \\ q \neq p}}^{c} \sum^{c} m_{i,p,k} \cdot m_{i,q,k} \tag{10}$$

$$V_{\text{class}}^k = \frac{1}{m}\frac{1}{c} \sum_{i=1}^{m} \sum_{p=1}^{c} \left(M_{i,p,k} - \frac{1}{c} \sum_{q=1}^{c} M_{i,q,k} \right)^2. \tag{11}$$

Similarly, to preserve the diversity of features while reducing repetitive information at the timestep dimension different from the class dimension, the timestep-oriented significance score τ_t^k at index k is designed to be calculated as

$$\tau_t^k = -\beta U_t^k + (1-\beta) V_t^k \tag{12}$$

where $\beta \in (0, 1)$, U_t^k denotes the inter-timestep similarity at index k obtained by summing the average cosine similarities across timesteps at all the classes, and V_t^k is the inter-timestep variance at index k which is the sum of the variances across classes at all the timesteps, defined as follows:

$$U_t^k = \frac{1}{c}\frac{1}{m^2} \sum_{i=1}^{c} \sum_{p=1}^{m} \sum_{\substack{q=1 \\ q \neq p}}^{m} M_{p,i,k} \cdot M_{q,i,k} \tag{13}$$

$$V_t^k = \frac{1}{C}\frac{1}{m} \sum_{i=1}^{c} \sum_{p=1}^{m} \left(m_{p,i,k} - \frac{1}{m} \sum_{q=1}^{m} m_{q,i,k} \right)^2. \tag{14}$$

Finally, the class and timestep-oriented significance score of each index $k \in \{1, \ldots, d\}$ is obtained as the final measurement

$$\tau^k = \tau_{\text{class}}^k + \tau_t^k. \tag{15}$$

We rank the indexes by the class and timestep-oriented significance score τ^k and generate the purification index by reserving the indexes of the top K highest τ^k, which indicates the most inter-class and inter-timestep divergence and discrimination. Then, multi-stage features in the timestep-wise center feature bank $\mathcal{B}_c$ multiply the purification index to obtain the purified timestep-wise center feature bank $\mathcal{B}_c'$.

D. Selective Timestep Feature Fusion

For each labeled HSI patch x_0', multi-timestep features contain textual information and contextual semantics from shallow to deep timesteps. Meanwhile, HSI data have different spectral characteristics for different HSI datasets, resulting in different spectral representation laws of timestep-wise features. Compared to the manual selection of single timestep features in previous works, we explore the multi-timestep features and perform selective timestep feature fusion to adaptively select different timestep features integrating textual information and contextual semantics and learn comprehensive multi-timestep multi-stage selective representations. The whole process is illustrated in Fig. 2.

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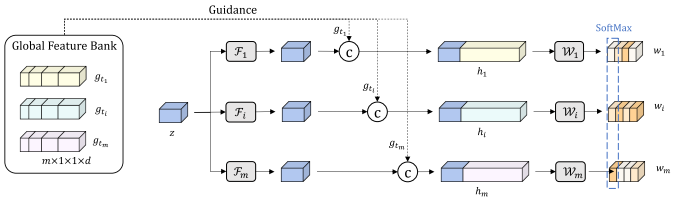


Fig. 4. Structure of global feature-guided selective timestep network in our proposed selective timestep feature fusion.

Specifically, for the purified timestep-wise center feature bank $\mathcal{B}'_c = \{c'_{t_i} | i \in \{1, \ldots, m\}\}$, our goal is to adaptively select the timestep of features in it for modeling spectral–spatial relations better. We first fuse all purified center features in $\mathcal{B}'_c$ to obtain the all-timestep feature $c_t \in \mathcal{R}^{1\times1\times K}$ via an element-wise summation

$$c_t = \sum_{i=1}^{m} c'_{t_i}. \tag{16}$$

Then, the all-timestep feature c_t is fed into a simple multi-layer perception (MLP) to obtain a compact feature $z \in \mathcal{R}^{1\times1\times K_r}$ with fewer channels for better efficiency

$$z = \mathcal{F}_{\text{mlp}}(c_t) = W_1(\delta(\text{BN}(W_2 c_t))) \tag{17}$$

where δ is the ReLU function, BN is the batch normalization, $W_1 \in \mathcal{R}^{K\times K_r}$, $W_2 \in \mathcal{R}^{K_r\times K_r}$, $K_r = K/2$.

However, adaptive timestep selection solely based on timestep-wise center features leads to imprecise choices due to the absence of spatial information from neighboring pixels around the central pixel. Hence, we propose a global feature-guided selective timestep network, designed to incorporate global features for guiding multi-timestep feature selection, thereby enriching spatial information in modeling spatial distributions. And the network is depicted in Fig. 4. Specifically, given a timestep-wise global feature bank $\mathcal{B}_g = \{g_{t_i} | i \in \{1, \ldots, m\}\}$, for each purified center feature c'_{t_i}, the corresponding global feature g_{t_i} is concatenated with z with the transformation $\mathcal{F}_i$ to obtain h_i. Then the linear projection $\mathcal{W}_i$ is applied to the h_i, respectively, $i \in \{1, \ldots, m\}$. To adaptively select different timesteps of center features, a softmax operator on the channel-wise digits is used to obtain the selective weights $\{w_i\}_{i=1}^m$ guided by the compact feature z and the corresponding global information

$$w'_i = \mathcal{W}(h_i) = \mathcal{W}\left(\left[\mathcal{F}_i(z), g_{t_i}\right]\right) \tag{18}$$

$$w_i^c = \frac{e^{w_i'^c}}{\sum_{j=1}^{m} e^{w_j'^c}} \tag{19}$$

where $\{\mathcal{F}_i\}_{i=1}^m$ and $\{\mathcal{W}_i\}_{i=1}^m$ are linear projections to align the channel dimension to K channels, and w_i^c is the cth element of the selective weight w_i, $c \in \{1, \ldots, K\}$. Finally, the multi-timestep multi-stage selective representation r_s is selected and fused through the selective weights $\{w_i\}_{i=1}^m$ on each purified center feature c'_{t_i}

$$r_s = \sum_{i=1}^{m} w_i c'_{t_i}. \tag{20}$$

After obtaining the multi-timestep multi-stage selective representation, a lightweight network is needed to predict the classification label. Inspired by Zhang et al. [45], we train an ensemble of lightweight linear classifiers that takes the dynamic pixel representations as inputs and predicts the classification label of each pixel. Specifically, each classifier is trained independently, consisting of two hidden layers with ReLU activation and batch normalization. When testing a sample, the final predicted label is obtained by majority voting of the ensemble of pixel classifiers, as illustrated in Fig. 2. This method brings more stability of prediction with a very small cost since the parameters of each classifier are very limited.

IV. Experiments and Results

In this section, we first describe four well-known HSI datasets, including the Indian Pines dataset, the Pavia University dataset, the Houston 2018 dataset, and the WHU-Hi-Longkou dataset. The experimental setting is then introduced including evaluation metrics, a brief introduction of compared state-of-art methods, and implementation details. Then, we conduct quantitative experiments and ablation analysis to evaluate our proposed method.

A. Datasets Description

1) Indian Pines: The Indian Pines dataset was acquired in 1992 over an area of Indian pines in North-Western Indiana by Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor. It consists of 145 × 145 pixels with a spatial resolution of 20 m and 220 spectral bands in the wavelength range of 400–2500 nm. There are 200 bands retained for classification (1–103, 109–149, 164–219) after removing the bands affected by noise. The dataset contains 10 249 labeled pixels with 16 categories. We use 10% of the labeled samples for training and the rest for testing. The class name and the number of training and testing samples are listed in Table I.

2) Pavia University: The Pavia University (PaviaU) dataset was collected in 2003 by the reflective optics system imaging spectrometer (ROSIS-3) sensor over a part of the city of Pavia, Italy. The dataset consists of 610 × 340 pixels with a spatial resolution of 1.3 m and 115 spectral bands in the wavelength range of 430–860 nm. 103 out of 115 bands are used for classification after removing 12 noisy bands. The image contains a large number of background pixels, and only 42 776 labeled pixels are divided into nine classes, including asphalt, meadows, gravel, and so on. We use 5% of the labeled samples for training and the rest for testing. The class name and the number of training and testing samples are listed in Table II.

3) Houston 2018: The Houston 2018 dataset, identified as the 2018 IEEE GRSS DFC dataset, was gathered in 2018 by the National Center for Airborne Laser Mapping (NCALM) over the University of Houston campus and its neighboring urban area, including HSI, multi-spectral LiDAR, and very high-resolution RGB images. The HSI dataset consists of 601 × 2384 pixels with a spatial resolution of 1 m and 48 spectral bands in the wavelength range of 380–1050 nm. It contains 504 856 labeled pixels and 20 classes of interest.

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TABLE I
LAND-COVER TYPES, THE NUMBER OF LABELED TRAINING SAMPLES, AND TESTING SAMPLES OF THE INDIAN PINES DATASET

Class	Land Cover Type	Training	Testing
1	Alfalfa	5	41
2	Corn-Notill	143	1285
3	Corn-Mintill	83	747
4	Corn	24	213
5	Grass-Pasture	48	435
6	Grass-Trees	73	657
7	Grass-Pasture-Mowed	3	25
8	Hay-Windrowed	48	430
9	Oats	2	18
10	Soybean-Notill	97	875
11	Soybean-Mintill	245	2210
12	Soybean-Clean	59	534
13	Wheat	20	185
14	Woods	126	1139
15	Buildings-Grass-Trees-Drives	39	347
16	Stone-Steel-Towers	9	84
	Total	1024	9225

TABLE II
LAND-COVER TYPES, THE NUMBER OF LABELED TRAINING SAMPLES, AND TESTING SAMPLES OF THE PAVIAU DATASET

Class	Land Cover Type	Training	Testing
1	Asphalt	332	6299
2	Meadows	932	17717
3	Gravel	105	1994
4	Trees	153	2911
5	Painted Metal Sheets	67	1278
6	Bare Soil	251	4778
7	Bitumen	67	1263
8	Self-Blocking Bricks	184	3498
9	Shadows	47	900
	Total	2138	40638

We use 5% of the labeled samples for training and the rest for testing. The class name and the number of training and testing samples are listed in Table III.

4) WHU-Hi-Longkou: The WHU-Hi-Longkou dataset was acquired in 2018 by an 8-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJ-innovations Matrice 600 Pro UAV platform. It consists of 550 × 400 pixels with a spatial resolution of 0.463 m and 270 spectral bands in the wavelength range of 400–1000 nm. It contains 204 542 labeled samples and nine object classes. We use 0.5% of the labeled samples for training and the rest for testing. The class name and the number of training and testing samples are listed in Table IV.

B. Experimental Setting

1) Evaluation Metrics: We evaluate the performance of all methods by three widely used indexes: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ).

2) Comparison With State-of-the-Art Methods: To demonstrate the effectiveness of our proposed method, we compare our classification performance with several state-of-the-art approaches using the most effective setting for these methods.

1) The 2-D CNN [13] architecture contains three 2-D convolution blocks and a softmax layer. Each convolution block consists of a 2-D convolution layer, a BN layer, an avg-pooling layer, and a ReLU activation function.

TABLE III
LAND-COVER TYPES, THE NUMBER OF LABELED TRAINING SAMPLES, AND TESTING SAMPLES OF THE HOUSTON 2018 DATASET

Class	Land Cover Type	Training	Testing
1	Healthy Grass	490	9309
2	Stressed Grass	1625	30877
3	Artificial turf	34	650
4	Evergreen trees	680	12915
5	Deciduous trees	251	4770
6	Bare earth	226	4290
7	Water	13	253
8	Residential buildings	1989	37783
9	Non-residential buildings	11187	212565
10	Roads	2293	43573
11	Sidewalks	1702	32327
12	Crosswalks	76	1442
13	Major thoroughfares	2317	44031
14	Highways	493	9372
15	Railways	347	6590
16	Paved parking lots	575	10925
17	Unpaved parking lots	7	139
18	Cars	327	6220
19	Trains	269	5100
20	Stadium seats	341	6483
	Total	25242	479614

TABLE IV
LAND-COVER TYPES, THE NUMBER OF LABELED TRAINING SAMPLES, AND TESTING SAMPLES OF THE WHU-HI-LONGKOU DATASET

Class	Land Cover Type	Training	Testing
1	Corn	172	34339
2	Cotton	42	8332
3	Sesame	15	3016
4	Broad-leaf soybean	316	62896
5	Narrow-leaf soybean	21	4130
6	Rice	59	11795
7	Water	335	66721
8	Roads and houses	36	7088
9	Mixed weed	26	5203
	Total	1022	203520

2) The 3-D CNN [13] contains three 3-D convolution blocks and a softmax layer. Each 3-D convolution block consists of a 3-D convolution layer, a BN layer, a ReLU activation function, and a 3-D convolution layer with step size 2.
3) The SSRN [14] is a spectral–spatial residual network based on 3-D CNN and residual connection. Spectral residual blocks and spatial residual blocks are designed to extract discriminative features from HSI data.
4) For SF [16], group-wise spectral embedding and cross-layer adaptive fusion modules in the transformer framework are adopted to capture local spectral representations from neighboring bands.
5) The SSFTT [17] systematically combines CNN network and transformer structure to exploit spectral–spatial information in the HSI, with a Gaussian weighted feature tokenizer module making the samples more separable.
6) The GAHT [18] is an end-to-end group-aware transformer method with three-stage hierarchical framework.
7) The 3DCAE [20] is an unsupervised method using an encoder–decoder backbone with 3-D convolution operation to learn spectral–spatial features.

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TABLE V

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, AND κ AS WELL AS THE ACCURACIES FOR EACH CLASS ON THE INDIAN PINES DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Class	2-D CNN	3-D CNN	SSRN	SF	SSFTT	GAHT	3DCAE	3DAES	UMSDFL	SpectralDiff	MTMSD
1	65.85	58.54	94.14	63.00	95.12	97.56	72.97	**100.00**	99.10	**100.00**	**100.00**
2	**99.77**	76.19	97.84	92.35	97.67	98.05	88.50	89.34	96.10	97.90	99.52
3	81.66	77.64	97.54	86.86	98.87	98.66	87.20	95.98	95.39	98.93	**99.01**
4	96.71	52.11	90.70	88.96	91.55	95.31	84.90	95.31	97.24	**100.00**	99.62
5	85.75	93.56	97.75	92.49	96.32	95.17	90.28	88.74	94.12	94.02	**98.62**
6	97.87	98.17	99.24	99.12	99.54	99.85	97.97	99.09	99.25	99.54	**99.97**
7	**100.00**	36.00	81.60	52.50	**100.00**	**100.00**	56.52	56.00	88.46	**100.00**	99.20
8	**100.00**	98.60	**100.00**	99.16	**100.00**	**100.00**	99.48	99.77	**100.00**	**100.00**	**100.00**
9	50.00	55.56	74.44	41.18	88.89	**100.00**	87.50	**100.00**	94.44	**100.00**	**100.00**
10	35.54	82.86	94.77	93.16	97.71	94.29	86.80	87.77	95.84	98.51	**98.79**
11	88.01	90.45	98.87	92.27	98.69	99.37	96.68	96.88	99.29	**99.77**	99.67
12	98.13	62.55	97.83	85.44	98.13	96.63	80.83	83.71	93.37	91.76	**98.84**
13	99.46	88.65	99.24	99.02	97.28	**100.00**	**100.00**	**100.00**	**100.00**	99.46	99.78
14	99.91	99.39	99.18	96.73	99.91	97.89	99.90	99.30	95.25	**99.91**	99.86
15	91.35	86.17	93.95	83.41	98.84	97.12	96.80	98.56	99.15	98.27	**99.42**
16	86.90	45.24	98.33	93.50	95.54	94.05	84.00	97.62	**100.00**	98.81	98.10
OA (%)	87.77	85.42	97.75	92.31	97.47	97.95	92.69	94.34	97.02	98.54	**99.45**
AA (%)	86.06	75.10	94.71	84.95	96.57	97.75	88.15	93.00	96.00	98.56	**99.40**
κ	0.8603	0.8324	0.9743	0.9124	0.9711	0.9766	0.9162	0.9353	0.9660	0.9833	**0.9937**

TABLE VI

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, AND κ, AS WELL AS THE ACCURACIES FOR EACH CLASS ON THE PAVIAU DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Class	2-D CNN	3-D CNN	SSRN	SF	SSFTT	GAHT	3DCAE	3DAES	UMSDFL	SpectralDiff	MTMSD
1	99.68	97.02	98.81	96.21	99.33	99.38	94.20	97.94	99.62	99.89	**100.00**
2	99.41	99.97	99.83	99.64	99.92	99.80	99.58	99.00	99.98	**100.00**	**100.00**
3	86.56	92.98	92.45	87.65	98.29	98.35	78.83	89.57	91.02	98.75	**100.00**
4	98.18	97.53	98.32	96.64	98.49	99.52	97.53	98.63	98.40	97.35	**99.59**
5	99.84	99.06	99.65	99.97	99.53	**100.00**	**100.00**	**100.00**	**100.00**	95.93	**100.00**
6	**100.00**	99.10	99.43	99.56	**100.00**	99.75	95.30	97.11	98.32	**100.00**	**100.00**
7	99.84	79.10	99.76	90.30	99.13	99.60	97.42	94.77	99.61	**100.00**	**100.00**
8	**100.00**	97.34	99.32	94.60	98.05	98.63	96.87	98.20	98.36	99.77	99.87
9	98.22	95.22	99.82	98.49	95.44	99.33	98.61	97.44	**99.89**	94.44	99.79
OA (%)	98.86	97.88	99.10	97.54	99.21	99.53	96.77	97.92	99.02	99.46	**99.95**
AA (%)	97.97	95.26	98.60	95.88	98.69	99.37	95.37	96.96	98.36	98.46	**99.92**
κ	0.9848	0.9719	0.9881	0.9674	0.9915	0.9937	0.9571	0.9725	0.9870	0.9929	**0.9994**

8) The 3DAES [46] is a semi-supervised method using an autoencoder to extract spectral–spatial features from unlabeled samples and then optimizes the Siamese network and classifier using constructed sample pairs.
9) The UMSDFL [21] is an unsupervised method using encoder and decoder with convolutional layers to learn spectral–spatial features. A clustering branch and a multi-layer fusion module are designed to enhance the features.
10) The SpectralDiff [32] is a diffusion-based unsupervised method that learns diffusion features through the spectral–spatial diffusion module and feeds diffusion features into the attention-based classification module.

3) Implementation Details: The proposed MTMSD was implemented using the Pytorch framework. The patch size is set to 48 × 48, and the dimension of PCA is set to $8/N$, where N is the number of spectral bands of the dataset. In the diffusion-pretraining procedure, we use Kullback–Leibler divergence loss as the loss function. The Adam optimizer is adopted with a batch size of 128 and a learning rate of 1e−4, training a total of 40k steps. In the feature-exploring stage, the cross-entropy loss is used in lightweight classifiers. m and K are set to be 19 and 5, respectively. We adopt the Adam optimizer and the cosine annealing as our training schedule. The original learning rate and minimum learning rate are set to be 1e−4 and 5e−6, respectively. The number of epochs is set to 100 for all datasets. We calculate the results fairly by averaging the results of ten repeated experiments with different training sample selections.

C. Quantitative Results and Analysis

1) Classification Results Compared With SOTA Methods: Quantitative classification results in terms of class-specific accuracy, OA, AA, and κ of the compared methods on the Indian Pines, PaviaU, Houston 2018, and Longkou datasets are listed in Tables V–VIII, respectively. The classification maps of all methods are shown in Figs. 5–8.

Compared with other methods, our proposed MTMSD achieves the highest OA, AA, and κ on four datasets. According to the results, the CNN-based supervised methods, 2-D CNN, 3-D CNN, and SSRN, obtain good performance owing to their ability to capture local spatial information. Besides,

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TABLE VII

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, AND κ AS WELL AS THE ACCURACIES FOR EACH CLASS ON THE HOUSTON 2018 DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Class	2-D CNN	3-D CNN	SSRN	SF	SSFTT	GAHT	3DCAE	3DAES	UMSDFL	SpectralDiff	MTMSD
1	87.12	82.02	86.30	**92.36**	79.93	79.50	91.60	81.80	88.99	83.47	88.87
2	92.05	96.64	95.32	95.08	93.44	96.55	93.92	93.30	**97.53**	92.48	96.29
3	96.92	96.00	99.72	96.21	99.66	**100.00**	97.60	89.23	99.23	99.38	99.93
4	98.05	95.55	97.49	98.48	96.64	97.62	95.92	94.39	97.98	97.01	**99.32**
5	87.25	79.16	86.09	91.87	90.11	95.01	85.79	81.95	92.51	86.25	**97.01**
6	95.36	97.51	98.48	99.62	99.62	99.91	98.50	97.72	99.58	99.58	**100.00**
7	96.44	71.94	93.68	27.01	85.45	95.65	61.15	83.40	96.85	87.35	**97.40**
8	97.15	88.62	91.62	95.42	98.72	99.18	91.15	91.27	94.20	98.73	**99.81**
9	98.24	92.80	97.88	98.60	99.09	99.35	95.12	97.12	99.07	99.31	**99.72**
10	93.88	71.61	81.72	88.22	91.15	92.64	76.72	76.90	88.61	87.73	**96.63**
11	75.85	73.17	69.44	27.01	80.97	85.73	70.22	71.14	82.28	79.33	**93.96**
12	12.55	11.10	0.51	31.46	41.69	33.43	4.79	3.19	42.76	37.24	**55.30**
13	85.24	69.12	84.59	91.79	94.49	96.79	90.52	84.62	93.26	95.64	**97.87**
14	77.12	96.18	89.65	92.91	96.97	99.17	87.23	95.41	97.02	98.09	**99.27**
15	94.45	98.98	99.34	99.33	99.30	**99.92**	99.06	97.31	99.59	99.47	**99.92**
16	93.43	90.40	91.00	96.36	97.84	98.15	92.01	87.79	96.36	98.42	**99.84**
17	64.75	20.86	0.00	22.78	69.21	90.65	0.00	46.04	**100.00**	84.17	**100.00**
18	91.70	89.05	93.66	91.61	93.07	97.85	90.43	83.89	94.24	93.91	**99.60**
19	96.88	95.45	96.92	96.53	97.97	99.84	96.09	94.55	99.59	98.47	**99.99**
20	99.83	93.92	99.12	99.77	99.96	**100.00**	97.27	96.30	99.89	**100.00**	**100.00**
OA (%)	93.38	86.88	91.61	90.65	95.48	96.69	90.34	90.39	95.38	95.28	**98.29**
AA (%)	86.71	80.50	82.63	80.75	90.26	92.85	80.76	82.37	92.98	90.80	**96.04**
κ	0.9137	0.8313	0.8906	0.8784	0.9412	0.9570	0.8751	0.8748	0.9398	0.9385	**0.9777**

TABLE VIII

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, AND κ, AS WELL AS THE ACCURACIES FOR EACH CLASS ON THE WHU-HI-LONGKOU DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Class	2-D CNN	3-D CNN	SSRN	SF	SSFTT	GAHT	3DCAE	3DAES	UMSDFL	SpectralDiff	MTMSD
1	99.44	99.55	99.80	99.76	99.85	99.91	99.71	98.85	99.93	99.53	**99.97**
2	95.44	93.18	98.82	89.20	95.96	98.99	95.07	98.78	97.40	97.49	**99.67**
3	81.76	93.83	91.11	97.25	93.50	96.68	85.18	80.64	91.25	**100.00**	97.67
4	**99.99**	98.70	99.71	98.42	99.04	99.55	98.26	99.71	99.13	99.46	99.81
5	75.45	83.74	94.04	83.80	92.42	95.96	60.39	84.02	94.19	96.39	**97.31**
6	98.66	98.88	99.89	97.69	99.34	99.77	99.49	98.76	99.76	99.33	**99.86**
7	99.96	99.99	99.97	99.98	99.99	99.99	99.99	99.94	99.99	99.85	**99.99**
8	**98.41**	96.66	96.33	89.45	97.19	96.46	96.67	95.40	97.95	86.82	96.77
9	91.08	94.77	93.95	73.44	96.16	86.51	92.89	89.54	88.93	87.28	**96.24**
OA (%)	98.58	98.48	99.28	97.47	99.02	99.19	97.86	98.54	98.99	98.71	**99.61**
AA (%)	93.36	95.37	97.07	92.11	97.05	97.09	91.96	93.96	96.20	96.24	**98.59**
κ	0.9812	0.9801	0.9905	0.9668	0.9872	0.9893	0.9718	0.9807	0.9868	0.9830	**0.9949**

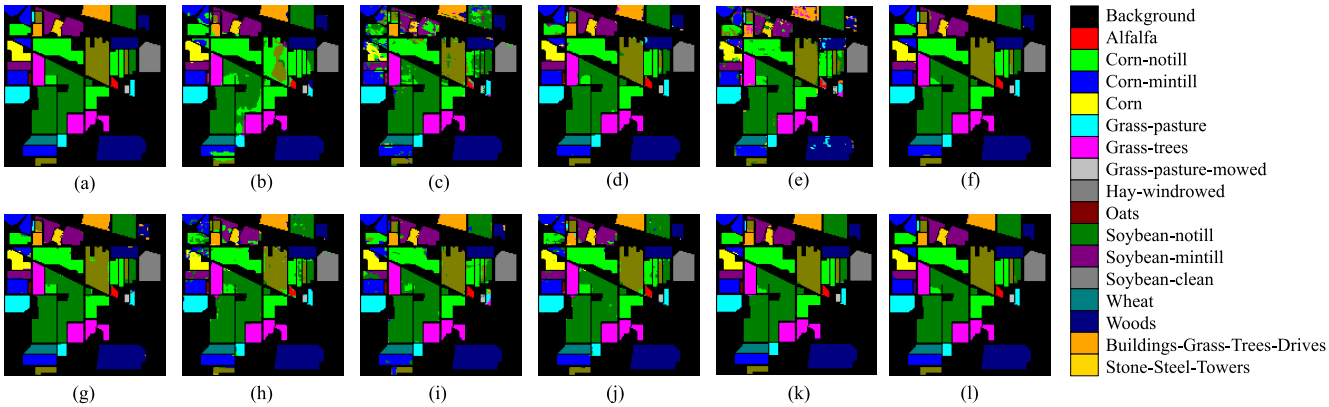


Fig. 5. Classification maps obtained by different methods on the Indian Pines dataset. (a) Ground truth. (b) 2-D CNN (OA = 87.77%). (c) 3-D CNN (OA = 85.42%). (d) SSRN (OA = 97.75%). (e) SF (OA = 92.31%). (f) SSFTT (OA = 97.47%). (g) GAHT (OA = 97.95%). (h) 3DCAE (OA = 92.69%). (i) 3DAES (OA = 94.34%). (j) UMSDFL (OA = 97.02%). (k) SpectralDiff (OA = 98.54%). (l) MTMSD (OA = 99.45%).

since transformers are capable of capturing sequential information, transformer-based supervised methods, SF, SSFTT, and GAHT, also achieve competitive performance. Unsupervised methods, 3DCAE, 3DAES, and UMSDFL, are proposed to tackle the problem of limited samples by learning representative features without any labeled samples. Limited by the

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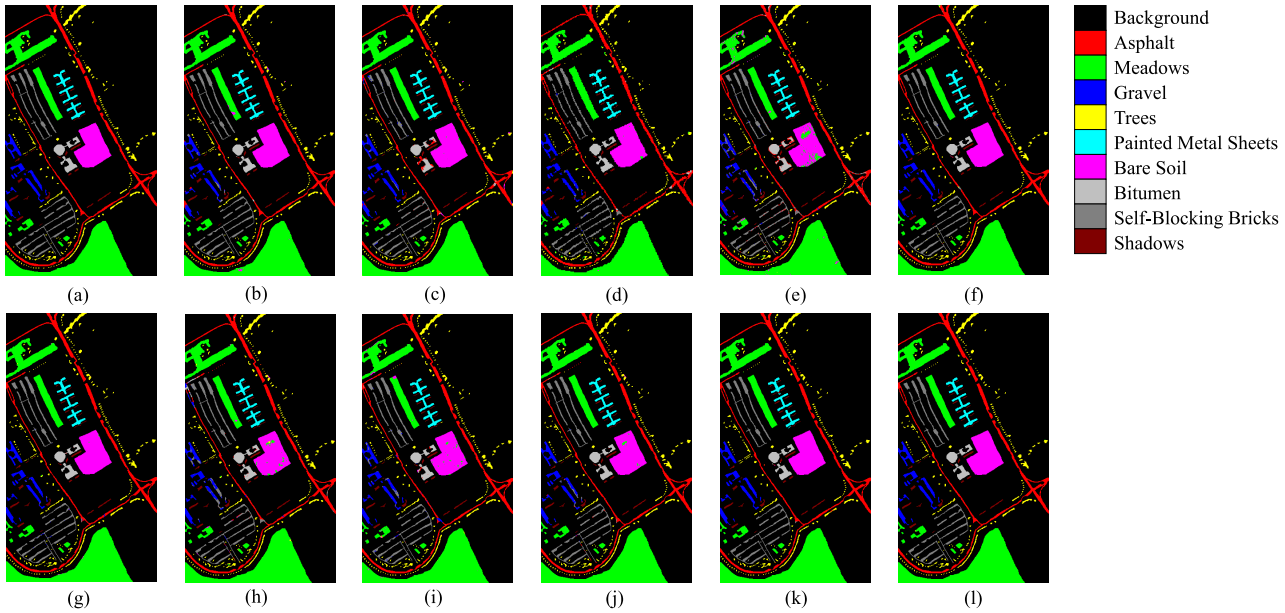


Fig. 6. Classification maps obtained by different methods on the PaviaU dataset. (a) Ground truth. (b) 2-D CNN (OA = 98.86%). (c) 3-D CNN (OA = 97.88%). (d) SSRN (OA = 99.10%). (e) SF (OA = 97.54%). (f) SSFTT (OA = 99.21%). (g) GAHT (OA = 99.53%). (h) 3DCAE (OA = 96.77%). (i) 3DAES (OA = 97.92%). (j) UMSDFL (OA = 99.02%). (k) SpectralDiff (OA = 99.46%). (l) MTMSD (OA = 99.95%).

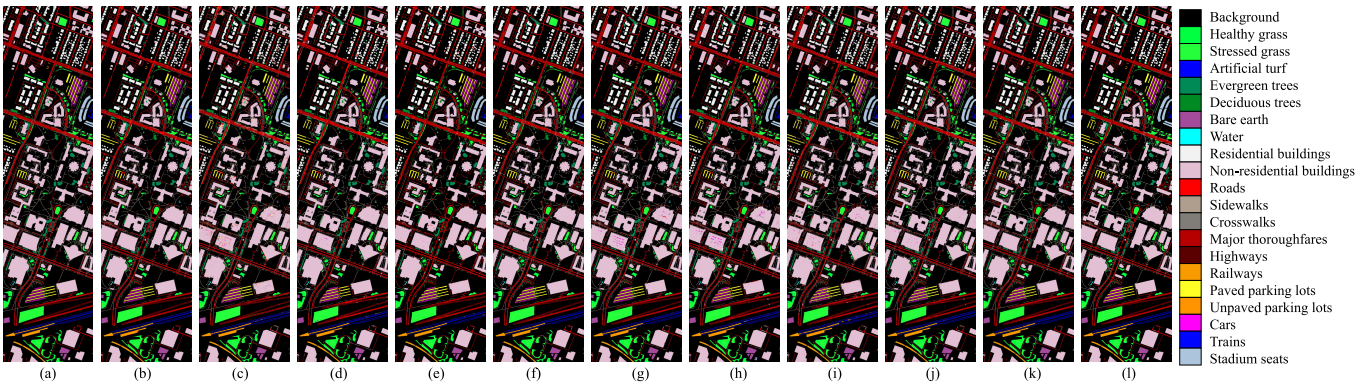


Fig. 7. Classification maps obtained by different methods on the Houston 2018 dataset. (a) Ground truth. (b) 2-D CNN (OA = 94.84%). (c) 3-D CNN (OA = 86.88%). (d) SSRN (OA = 91.61%). (e) SF (OA = 90.65%). (f) SSFTT (OA = 95.48%). (g) GAHT (OA = 96.69%). (h) 3DCAE (OA = 90.34%). (i) 3DAES (OA = 90.39%). (j) UMSDFL (OA = 95.38%). (k) SpectralDiff (OA = 95.28%). (l) MTMSD (OA = 98.29%).

model architecture, the features learned in an unsupervised manner are not discriminative enough for HSI classification lacking high-level information. Therefore, the performance of 3DCAE and 3DAES is even lower than that of some explicit learning methods. SpetralDiff introduces the diffusion model to HSI classification and achieves competitive performance due to the advantages of diffusion features. However, the feature used in the method is from a single timestep and a single layer with the loss of some key information, which limits the performance. Our proposed MTMSD explores the value of multi-timestep multi-stage diffusion features through CTMSFP and selective timestep feature fusion, effectively modeling complex spectral–spatial relations due to the adaptive integration of contextual semantics and textual details. Thus, our MTMSD outperforms all the previous methods on four datasets: Indian Pines, PaviaU, Houston 2018, and Longkou. Notably, the classification performance of the Houston 2018 Dataset is largely improved compared with the previous SOTA method in terms of OA (98.29% versus 96.69%), AA (96.04% versus 92.98%), and κ (0.9777 versus 0.9570), which especially demonstrate our effectiveness.

2) Classification Results With Different Proportions of Training Samples: The classification results with different proportions of training samples are shown in Fig. 9. It can be observed that the performance increases with the percentages of training samples. Our method outperforms the compared method consistently in terms of OA on four datasets. Especially, using only 2% of the training sample, our method achieves comparable results to other methods using 5% of the training sample on the Houston 2018 dataset.

D. Ablation Studies

In this section, we analyze the effect of the components in our method.

1) Ablation for CTMSFP: Table IX presents the effect of CTMSFP across the four datasets. The multi-stage features

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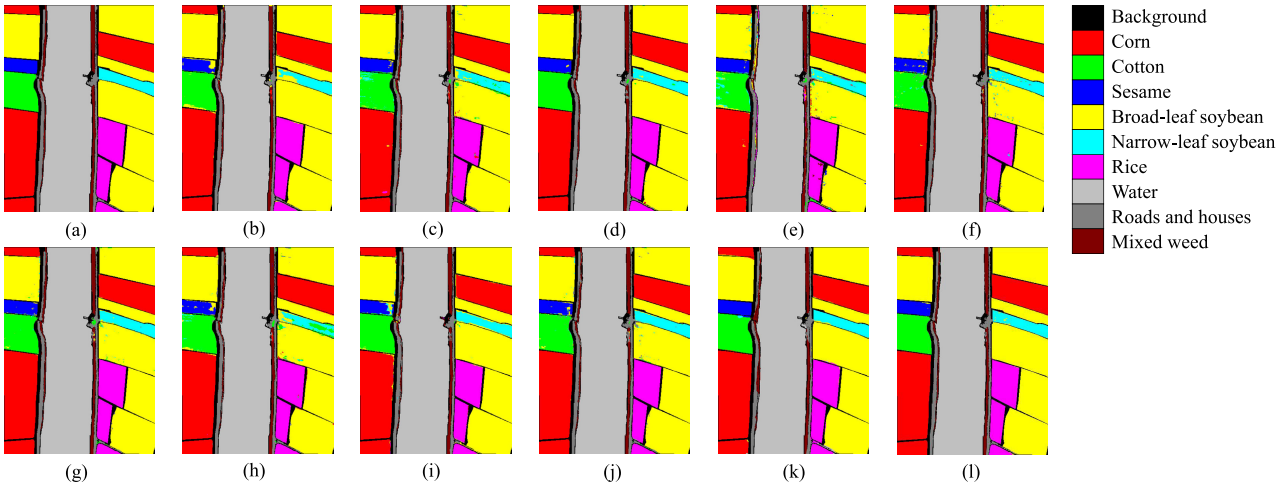


Fig. 8. Classification maps obtained by different methods on the WHU-Hi-Longkou dataset. (a) Ground truth. (b) 2-D CNN (OA = 98.58%). (c) 3-D CNN (OA = 98.48%). (d) SSRN (OA = 99.28%). (e) SF (OA = 97.47%). (f) SSFTT (OA = 99.02%). (g) GAHT (OA = 99.19%). (h) 3DCAE (OA = 97.86%). (i) 3DAES (OA = 98.54%). (j) UMSDFL (OA = 98.99%). (k) SpectralDiff (OA = 98.71%). (l) MTMSD (OA = 99.61%).

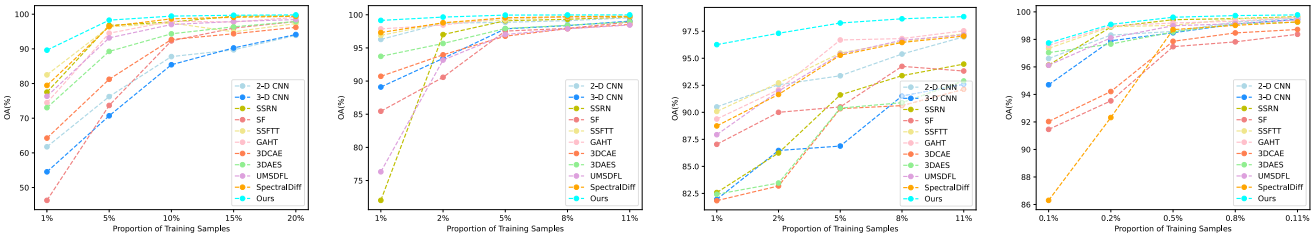


Fig. 9. Classification performance of the compared methods with different proportions of training samples on four datasets. (a) Indian Pines. (b) PaviaU. (c) Houston 2018. (d) Longkou.

TABLE IX

ABLATION FOR CTMSFP ON FOUR DATASETS IN TERMS OF OA, IT, PARAMETERS, AND GPU MEMORY. THE BEST RESULTS ARE SHOWN IN BOLD

Method	Indian Pines		PaviaU		Houston 2018		Longkou		Param. (M)	GPU memory (G)
	OA (%)	IT (s)	OA (%)	IT (s)	OA (%)	IT (s)	OA (%)	IT (s)		
SpectralDiff	98.54	10.92	99.46	22.41	95.28	268.88	98.71	178.25	**1.38**	3.09
MTMSD (w/o CTMSFP)	99.39	4.55	99.90	22.78	98.20	215.20	99.51	85.14	55.91	1.28
MTMSD	**99.45**	**3.52**	**99.95**	**12.41**	**98.29**	**168.52**	**99.61**	**61.53**	20.18	**0.46**

extracted from the diffusion model are not entirely aligned with the HSI classification task. Therefore, the CTMSFP is proposed to reduce the redundancy of the features and maintain efficiency. All experiments are conducted on an RTX 3090 GPU, with a consistent batch size of 64 used during training. As shown in Table IX, the results demonstrate that CTMSFP significantly reduces the number of parameters and the GPU memory consumption during training. This reduction is due to the channel-wise purification performed by CTMSFP before the features are input into the model, which substantially decreases the size of both the input tensor and the model structure. Furthermore, the average inference time (IT) is reduced by 25%, effectively enhancing the model's computational efficiency, while still ensuring a slight performance improvement. We also compare the performance and efficiency of another diffusion-based HSI classification model, SpectralDiff, which utilizes single-timestep single-stage diffusion features. Our method performs independent linear transformations on features from different timesteps, allowing tailored transformations for diverse multi-timestep feature patterns, resulting in a larger number of parameters. However, our method with CTMSFP significantly leads to performance, inference speed, and GPU memory consumption.

2) Ablation for Selective Timestep Feature Fusion: The ablation results for the selective timestep feature fusion are shown in Table X. In the manual selection method, we extract features from a single timestep for classification. The optimal timestep is selected for each dataset, and the best result is shown in the table. However, using the features from only one single timestep leads to the loss of abundant spectral–spatial information. Although the optimal timestep is chosen for each dataset, it lacks adequate information, only containing textural features or semantics to model spectral–spatial relations, and is not flexible enough to accommodate different patch data, both of which limit performance. The average fusion method considers features from different timestep t, obtaining

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TABLE X
OA (%) OF THE PROPOSED MTMSD WITH DIFFERENT FEATURE FUSION AND GLOBAL-FEATURE GUIDANCE ON FOUR DATASETS. THE BEST RESULTS ARE SHOWN IN BOLD

Method	Guidance	Indian Pines	PaviaU	Houston 2018	Longkou
Manual Selection	✗	98.17	99.40	94.57	99.08
Average Fusion	✗	99.03	99.78	97.85	99.39
Selective Fusion	✗	99.31	99.90	98.07	99.53
	✓	**99.45**	**99.95**	**98.29**	**99.61**

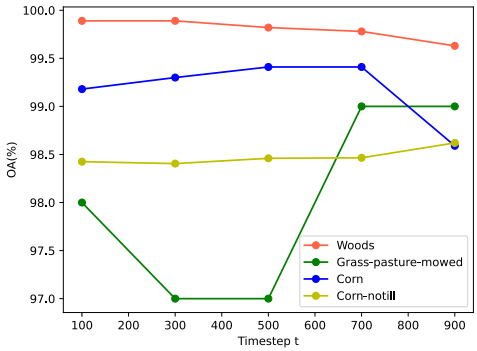


Fig. 10. Variation of classification results for four classes with the change of timestep on the Indian Pines Dataset.

better results than manual selection. However, assigning a uniform selection weight to features across all the timesteps indiscriminately does not fully harness their potential because the significance of features from different t for different data instances is heterogeneous. Therefore, the proposed selective fusion is more optimal for feature fusion. Compared with the average fusion, our selective timestep feature fusion achieves better performance by assigning the vital timestep features higher selecting weights. Furthermore, compared with no global-feature guidance, our proposed selective timestep fusion with global-feature guidance improves further due to the supplement of global information that enhances the ability to represent spatial distributions.

E. Discussion and Visualization

1) Analysis of Features From Multiple Timesteps: To analyze the features extracted from different timestep t, we record the change of the classification performance when changing t. For easy understanding, we choose four of the 16 classes to show their changes, as illustrated in Fig. 10. The performance for each class behaves differently as t increases. Features of larger t are more sensitive to class "corn-notill" and features of smaller t are more informative to class "woods." For class "Grass-pasture-mowed" and class "corn," features extracted at the intermediate t are the most discriminative. Thus, an appropriate fusion of features at different t is vital for accurate performance.

2) Visualization of Multi-Timestep Multi-Stage Diffusion Feature Exploration: The feature map of retained and removed channels after class and timestep-oriented multi-stage purification is visualized in Fig. 11. The retained channels capture more semantic information compared to the removed ones after purification, as evident from Fig. 11. Additionally, for the same channel, features at different timesteps exhibit diversity.

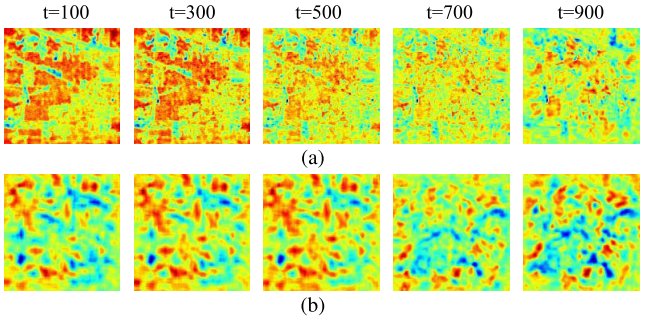


Fig. 11. Feature visualization of retained and removed channels after class and timestep-oriented multi-stage purification at different timestep t. (a) Retained. (b) Removed.

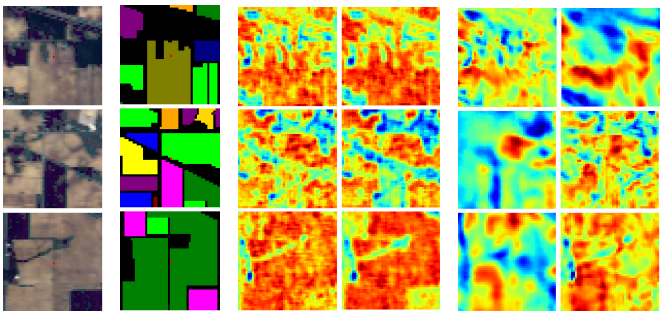


Fig. 12. Feature visualization of different timestep t with higher and lower selection weights. (a) Pseudocolor images. (b) Ground truth. (c) Higher weights. (d) Lower weights.

Features at shallow timesteps tend to capture stochastic details, while those at larger timesteps focus more on higher-level semantic information.

To further demonstrate the selective timestep feature fusion's ability to select features at different timesteps, we visualized the features with higher and lower selection weights for the same sample. As shown in Fig. 12, the results indicate that for features with higher selection weights, the response of the target pixels is more consistent with surrounding pixels of the same class, containing more classification-correlated information. If the features across all timesteps are averaged without discrimination, the important information relevant to the classification task may become obfuscated. Thus, to obtain an effective representation suitable for HSI classification, our proposed selective timestep feature fusion increases the proportion of classification-related information in the representation by assigning higher weights to more discriminative features. The visualization provides evidence that our proposed selective timestep feature fusion can effectively select timestep features suitable for classification tasks.

F. Parameter Analysis

In this section, we analyze the effect of various parameters that influence classification performance by training our proposed MTMSD in the same experimental setting as Section IV-B with different parameters.

1) Effect of Different Patch Sizes: First, we discuss the effect of different patch sizes on classification performance by two indexes, OA and AA. As shown in Fig. 13(a), the

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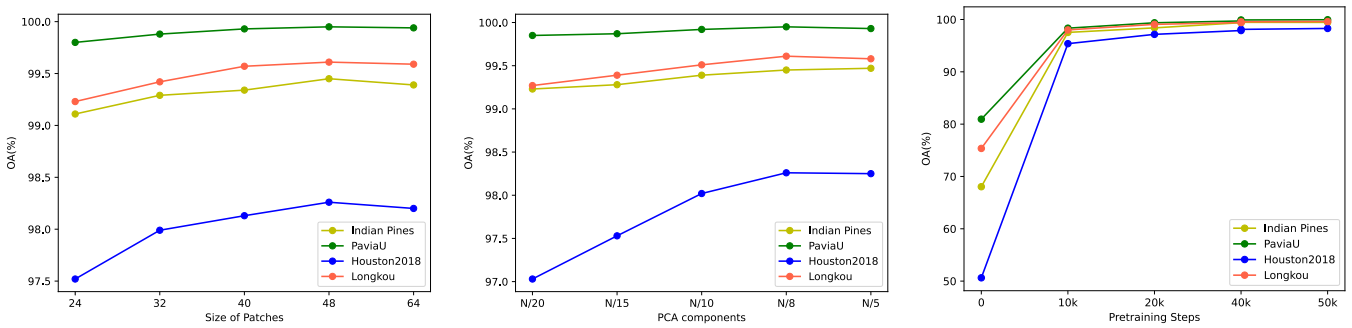


Fig. 13. Classification results of different parameters on the Indian Pines, PaviaU, Houston 2018, and Longkou Dataset. (a) Patch size. (b) PCA component. (c) Pretraining steps.

TABLE XI
GPU IT (S) OF DIFFERENT METHODS

Method	Indian Pines	PaviaU	Houston 2018	Longkou
2-D CNN	3.61	7.02	47.47	69.86
3-D CNN	3.83	7.66	48.99	65.24
SSRN	2.74	5.36	35.94	37.99
SF	3.52	7.59	63.73	48.40
SSFTT	3.79	7.28	31.38	59.32
GAHT	2.51	5.25	44.90	34.19
3DCAE	6.12	12.06	193.07	83.91
3DAES	2.81	5.64	41.64	41.83
UMSDFL	14.28	31.04	293.41	249.03
SpectralDiff	10.92	22.41	268.88	178.25
MTMSD (Ours)	3.52	12.41	168.52	61.53

patch size varies from 24 × 24 to 60 × 60. As the patch size increases, the performance first increases and then decreases. The best performance is obtained when the patch size is 48 × 48 with OA of 99.39% and AA of 99.30%. Too small patches contain insufficient spatial information and too large patches reduce the attention to detailed structures. Thus, we choose 48 × 48 to be the patch size for the proposed MTMSD.

2) Effect of Different PCA Components: This section analyzes the influence of the number of PCA components, which determines how much spectral information is retained in the compressed data. As the number of PCA components increases, more spectral information is retained while more computational cost and more redundant information are brought. Since each dataset has a different number of channels, the range of PCA components is different for four datasets. Assuming that N is the channel number of a dataset, the number of PCA components varies from $N/20$ to $N/5$. According to the results shown in Fig. 13(b), the best performance is achieved at PCA components of $N/8$.

3) Effect of Different Pretraining Steps: We validate the effectiveness of pretraining on the four datasets in terms of OA. As shown in Fig. 13(c), only 10k steps pretraining brings dramatic improvement (more than 30% OA) to the final classification performance. Furthermore, as the pretraining steps increase, the performance continues to rise to the best at around 40k steps.

G. Efficiency Analysis

We evaluate the IT of different methods on four public datasets to analyze the efficiency. All the experiments are carried out on a Nvidia RTX 3090. As shown in Table XI, although our method is not the fastest in inference speed, it achieves the best classification performance since it utilizes effective multi-timestep multi-stage diffusion features. It is noted that our method has reduced the average IT by 62% across four datasets compared to another diffusion-based HSI classification method, SpectralDiff.

V. CONCLUSION

HSI contains rich spectral–spatial information and complex relations, which are critical for classification tasks. Many supervised and unsupervised deep learning methods are proposed to learn spectral–spatial features from HSI data, achieving promising results in HSI classification. Recently, diffusion models as powerful models in generation and reconstruction tasks have been applied to HSI classification in one recent work. However, the diffusion features used in the work are extracted solely from a single timestep and a single stage of the denoising U-Net manually selected for each dataset, which limits the performance. Thus, we propose a diffusion-based feature learning framework that explores multi-timestep multi-stage diffusion features for HSI classification for the first time, named MTMSD. Quantitative experiments on four HSI datasets demonstrate that our proposed MTMSD outperforms state-of-the-art supervised and unsupervised methods.

ACKNOWLEDGMENT

The computations in this research were performed using the CFFF platform of Fudan University, Shanghai, China.

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Jingyi Zhou (Student Member, IEEE) received the B.E. degree from the School of Information Science and Technology, Fudan University, Shanghai, China, in 2023, where she is currently pursuing the M.E. degree.

Her main research interests include computer vision, hyperspectral analysis, sentiment analysis, and depth estimation.

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Jiamu Sheng received the B.E. degree from the School of Information Science and Technology, Fudan University, Shanghai, China, in 2022, where he is currently pursuing the M.E. degree with the Academy for Engineering and Technology.

His main research interests include computer vision, image quality assessment, and hyperspectral analysis.



Peng Ye is currently pursuing the Ph.D. degree with Fudan University, Shanghai, China.

He has published papers in leading journals and conferences, including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, *International Journal of Computer Vision* (IJCV), IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Oral, NeurIPS, ACM Multimedia (ACM MM), IEEE International Conference on Multimedia & Expo (ICME) Best Student Paper, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY (TCSVT), and International Conference on Acoustics, Speech and Signal Processing (ICASSP) Oral. He serves as a reviewer for various journals and conferences, including PAMI, IJCV, TCSVT, CVPR, European Conference on Computer Vision (ECCV), IEEE International Conference on Computer Vision (ICCV), and Machine Intelligence Research (MIR). His research interests include computer vision, network design, and network optimization.



Jiayuan Fan (Member, IEEE) received the Ph.D. degree in information engineering from Nanyang Technological University, Singapore, in 2015.

After her graduation, she worked as a Research Scientist at the Institute for Infocomm Research, A*STAR, Singapore. She is currently an Associate Professor with the Academy for Engineering and Technology, Fudan University, Shanghai, China. Her main research interests include computer vision and image forensic analysis and application.



Tong He received the Ph.D. degree in computer science from the University of Adelaide, Adelaide SA, Australia, in 2020.

He is currently a Researcher at Shanghai AI Laboratory, Shanghai, China. His research interests include computer vision and machine learning.



Bin Wang (Senior Member, IEEE) received the B.S. degree in electronic engineering and the M.S. degree in communication and electronic systems from Xidian University, Xi'an, China, in 1985 and 1988, respectively, and the Ph.D. degree in system science from Kobe University, Kobe, Japan, in 1999.

After his graduation in 1988, he was with Xidian University as a Teacher. From 1999 to 2000, he was with the Communications Research Laboratory, Ministry of Posts and Telecommunications, Kobe, as a Research Fellow, working on magnetoencephalography signal processing and its application for brain science. Then, as a Senior Supervisor, he was with the Department of Etching, Tokyo Electron AT Ltd., Tokyo, Japan, from 2000 to 2002, dealing with the development of advanced process control systems for etching semiconductor equipment. Since September 2002, he has been with the Department of Electronic Engineering, Fudan University, Shanghai, China, where he is currently a Full Professor and the Leader of the Image and Intelligence Laboratory. He has published more than 150 scientific articles in important domestic and international periodicals. He is the holder of several patents. His main research interests include multispectral/hyperspectral image analysis, automatic target/object detection and recognition, pattern recognition, signal detection and estimation, and machine learning.

Dr. Wang is an Associate Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (IEEE JSTARS).



Tao Chen (Senior Member, IEEE) received the Ph.D. degree in information engineering from Nanyang Technological University, Singapore, in 2013.

He was a Research Scientist at the Institute for Infocomm Research, A*STAR, Singapore, from 2013 to 2017, and a Senior Scientist at the Huawei Singapore Research Center, Singapore, from 2017 to 2018. Since 2019, he joined Fudan University, Shanghai, China, and led a research team focusing on light deep vision model design, multimodal vision analysis, and edge device-aware vision applications. To date, he has undertaken multiple projects and fundings from various government agencies such as NSFC and corporations like Huawei, Tencent. He has published over 110 academic papers in various reputable journals and conferences like IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), *International Journal of Computer Vision* (IJCV), IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE Conference on Computer Vision and Pattern Recognition (CVPR), and NeurIPS, and has granted over ten Patent Cooperation Treaty (PCT) patents.

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